

Friction Stations

1 Book Reading

Use the reading from **Section 4.2: Friction (pp. 59-60)** in *Conceptual Physics* to answer these questions.

1. What does friction do? And what causes friction?

2. Explain what Figure 4.3 is attempting to show.

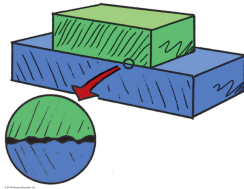


Figure 4.3

3. In which direction does the force of friction act?
4. As you push a crate harder and harder, what happens to the force of friction before the crate starts to move?
5. What's the difference between *sliding friction* and *static friction*? Which is stronger?

Name: _____ Date: _____ Period: _____

Name: _____ Date: _____ Period: _____

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6. Use your answer from question 5 to explain why antilock braking systems in cars prevent sliding on the ice.
7. How does the amount of friction depend on the speed that an object is moving?
8. How does the amount of friction depend on the area of contact between surfaces?
9. In the last paragraph on p. 60, the book discusses *air drag* as an example of fluid friction. How does air drag/fluid friction behave differently than the friction due to solids?

2 Spring Scales

1. Hook up the **10-N spring scale** to one of the blocks. Place 800 grams on top of the block. Pull the block until it moves. Pay particular attention to what happens to the reading on the scale just before it “budges.” Using what you learned in Section 1, explain how this shows the difference between static friction and sliding (kinetic) friction.

2. Write down the maximum value of the static friction that the scale reads just before the block “budges” and starts to move.

	Trial 1	Trial 2	Trial 3	Average
Maximum Static Friction				

3. To measure the sliding (kinetic) friction acting on the block, you need to pull the block across the table top at a constant velocity. While moving at a constant velocity, the scale will read the value for the kinetic friction. Do this three times and average the values together. Make sure to keep the 800 grams on top of the block

	Trial 1	Trial 2	Trial 3	Average
Kinetic (Sliding) Friction (<i>block is sitting flat</i>)				

4. Given what you learned in Section 1, what do you think will be the result of adding weight to the block? Will there be more friction, less friction, or the same amount of friction?

5. Let’s test your prediction. Remove all of the weight from the block and place just one single 200-g weight on the block. Measure *sliding (kinetic) friction*. Repeat with 400 g and 600 g.

Wood on table	Trial 1	Trial 2	Trial 3	Average
Block + 200 g				
Block + 400 g				
Block + 600 g				

6. Evaluate your prediction from question 4. Were you correct?

Name: _____

Date: _____

Period: _____

7. What do you think will happen if you turn the block upside down so that the rubber side is facing down? Will you have more or less friction?

8. Let's test your prediction.

Rubber on table	Trial 1	Trial 2	Trial 3	Average
Block + 200 g				
Block + 400 g				
Block + 600 g				

9. Evaluate your prediction from question 7. Were you correct?

10. The wood block has two surfaces it can slide over. If you turn the block on its side you can measure the friction of a smaller surface area. Given what you learned in Section 1, what do you think will be the result of doing this? Will there be more friction, less friction, or the same amount of friction?

11. Let's test your prediction. You can copy your data for the first line from question 3. Then, turn the block on its side, make sure to stack the 800 grams on top, and measure the kinetic friction.

	Trial 1	Trial 2	Trial 3	Average
Kinetic (Sliding) Friction (<i>block is sitting flat</i>)				
Kinetic (Sliding) Friction (<i>block is on its side</i>)				

12. Evaluate your prediction from question 10. Were you correct?